

12.421/12.621 Lecture 6

Reflection and transmission of EM waves
at oblique incidence and multiple layers

Topics Covered

12.421/12.621 Lecture 5 Review

EM BCs

1. $\epsilon_1' \underline{E}_1^\perp = \epsilon_2' \underline{E}_2^\perp$ - or - $\hat{n} \cdot (\epsilon_1' \underline{E}_1 - \epsilon_2' \underline{E}_2) = 0$ norm. comp. of $\underline{E}$ is cont.
2. $\mu_1 \underline{H}_1^\perp = \mu_2 \underline{H}_2^\perp$ - or - $\hat{n} \cdot (\mu_1 \underline{H}_1 - \mu_2 \underline{H}_2) = 0$ norm. comp. of $\underline{H}$ is cont.
3. $\underline{E}_1^\parallel = \underline{E}_2^\parallel$ - or - $\hat{n} \times (\underline{E}_1 - \underline{E}_2) = 0$ tang comp of $\underline{E}$ is cont.
4. $\underline{H}_1^\parallel = \underline{H}_2^\parallel$ - or - $\hat{n} \times (\underline{H}_1 - \underline{H}_2) = 0$ tang comp of $\underline{H}$ is cont.

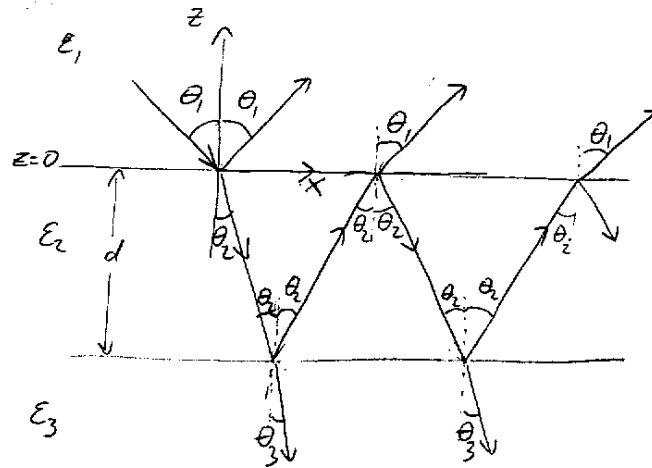
Laws of geometric optics

1. Incident, reflected, and transmitted rays and normal to interface are coplanar $\rightarrow$ plane of incidence
2. $\theta_i = \theta_r = \theta_t$
3. $\sqrt{\epsilon_2} \sin \theta_2 = \sqrt{\epsilon_1} \sin \theta_1$

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1) Reflection from a thin layer ($d \lesssim \delta_p$) overlying a half space

- Many cases where we may want to study thin-layer scattering
 E_g , Sea ice, snow on the ground, oil slicks on ocean, etc.
- Consider the case of homogeneous materials w/ planar interfaces



- If d is sufficiently small (define later, but intuition...), and
 - reflectivity at interface 12 is sufficiently small and, reflectivity at interface 23 is sufficiently large
 - we can
 $\rightarrow$ define an effective reflectivity of the two-layer composite in terms of total upward propagating E fields in medium 1
- Generally, can define reflectivity of N -layer composite by defining dielectric properties and applying BCs to $N-1$ interfaces
 - $\rightarrow$ can be constructed in a convenient matrix form for straightforward calculation of effective reflectivity of N -layer structure and transmission into bottom layer
 - $\rightarrow$ Here we consider only 3 layers pictured above

Ret from 2-layer comp (cont)

- Consider an H-polarized ^{incident} E-field

$$\underline{E}_1^- = \hat{y} A_1 e^{-\gamma_1 (x \sin \theta_1 - z \cos \theta_1)}$$

- where A_1 is the amplitude of the incident field (were calling this E_{0h}^i last class, simplified here)

- note that superscript - sign indicates downward propagating
- note use of numerical subscripts instead of i, e, r → avoid confusion
- note that $\gamma_1 \equiv j k_0 \sqrt{\epsilon_1}$

- Upward prop. wave in medium 1

$$\underline{E}_1^+ = \hat{y} B_1 e^{-\gamma_1 (x \sin \theta_1 + z \cos \theta_1)} = \hat{y} \rho A_1 e^{-\gamma_1 (x \sin \theta_1 + z \cos \theta_1)}$$

- where $B_1 \equiv$ amplitude of sum of all upward E fields

- note that superscript + indicates upward propagating

$$\rho = \frac{B_1}{A_1} \equiv \text{effective reflection coef}$$

- Total E field in medium 1 is

$$\underline{E}_1 = \underline{E}_1^- + \underline{E}_1^+ = \hat{y} A_1 \left(e^{\gamma_1 z \cos \theta_1} + \rho e^{-\gamma_1 z \cos \theta_1} \right) e^{-\gamma_1 x \sin \theta_1}$$

- We know that $\underline{H} = \frac{1}{\eta} (\hat{k} \times \underline{E})$

total
⇒ mag field in medium 1 is

$$\underline{H}_1 = \hat{x} H_{1x} + \hat{z} H_{1z}$$

$$H_{1x} = \frac{A_1 \cos \theta_1}{\eta_1} \left(e^{\gamma_1 z \cos \theta_1} - \rho e^{-\gamma_1 z \cos \theta_1} \right) e^{-\gamma_1 x \sin \theta_1}$$

$$H_{1z} = \frac{A_1 \sin \theta_1}{\eta_1} \left(e^{\gamma_1 z \cos \theta_1} + \rho e^{-\gamma_1 z \cos \theta_1} \right) e^{-\gamma_1 x \sin \theta_1}$$

- In medium 2, can follow the same procedure to get

$$\underline{E}_2 = \underline{E}_2^- + \underline{E}_2^+ = \hat{y} \left(A_2 e^{\gamma_2 z \cos \theta_2} + B_2 e^{-\gamma_2 z \cos \theta_2} \right) e^{-\gamma_2 x \sin \theta_2}$$

$$\underline{H}_2 = \hat{x} H_{2x} + \hat{z} H_{2z}$$

$$H_{2x} = \frac{\cos \theta_2}{\eta_2} \left(A_2 e^{\gamma_2 z \cos \theta_2} - B_2 e^{-\gamma_2 z \cos \theta_2} \right) e^{-\gamma_2 x \sin \theta_2}$$

$$H_{2z} = \frac{\sin \theta_2}{\eta_2} \left(A_2 e^{\gamma_2 z \cos \theta_2} + B_2 e^{-\gamma_2 z \cos \theta_2} \right) e^{-\gamma_2 x \sin \theta_2}$$

Rel from 2-layer comp (cont.)

- In medium 3, only downward prop

$$\underline{E}_3 = \hat{y} (A_3 e^{\gamma_3 z \cos \theta_3}) e^{-\gamma_3 x \sin \theta_3}$$

$$\underline{H}_3 = \hat{x} H_{3x} + \hat{z} H_{3z}$$

$$H_{3x} = \frac{\cos \theta_3}{\eta_3} A_3 e^{\gamma_3 z \cos \theta_3} e^{-\gamma_3 x \sin \theta_3}$$

$$H_{3z} = \frac{\sin \theta_3}{\eta_3} A_3 e^{\gamma_3 z \cos \theta_3} e^{-\gamma_3 x \sin \theta_3}$$

- Apply BCs

- Tangential E-field const

$$\Rightarrow \underline{E}_1|_{z=0} = \underline{E}_2|_{z=0}$$

$$\Rightarrow (A_1 + B_1) e^{-\gamma_1 x \sin \theta_1} = (A_2 + B_2) e^{-\gamma_2 x \sin \theta_2}$$

- Snell's Law:

$$\gamma_1 \sin \theta_1 = \gamma_2 \sin \theta_2 = \gamma_3 \sin \theta_3$$

$$\Rightarrow \boxed{A_1 + B_1 = A_2 + B_2}$$

- Tangent H-field

$$H_{1x}|_{z=0} = H_{2x}|_{z=0}$$

$$\Rightarrow \boxed{\frac{\cos \theta_1}{\eta_1} (A_1 - B_1) = \frac{\cos \theta_2}{\eta_2} (A_2 - B_2)}$$

- Similarly at 23 boundary

$$\underline{E}_2|_{z=-d} = \underline{E}_3|_{z=-d}$$

$$\text{and } H_{2x}|_{z=-d} = H_{3x}|_{z=-d}$$

$$\Rightarrow \boxed{\begin{aligned} A_2 e^{-\gamma_2 d \cos \theta_2} + B_2 e^{\gamma_2 d \cos \theta_2} &= A_3 e^{-\gamma_3 d \cos \theta_3} \\ \frac{\cos \theta_2}{\eta_2} (A_2 e^{-\gamma_2 d \cos \theta_2} - B_2 e^{\gamma_2 d \cos \theta_2}) &= \frac{A_3 \cos \theta_3}{\eta_3} e^{-\gamma_3 d \cos \theta_3} \end{aligned}}$$

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Ref from 2 layer comp (cont.)

- Assuming A_1 known, 4 Eqs. in 4 unknowns (B_1, A_2, B_2, A_3)
- Grinding through algebra gives

$$\rho = \frac{\rho_{12} + \rho_{23} e^{-2\gamma_2 d \cos \theta_2}}{1 + \rho_{12} \rho_{23} e^{-2\gamma_2 d \cos \theta_2}}$$

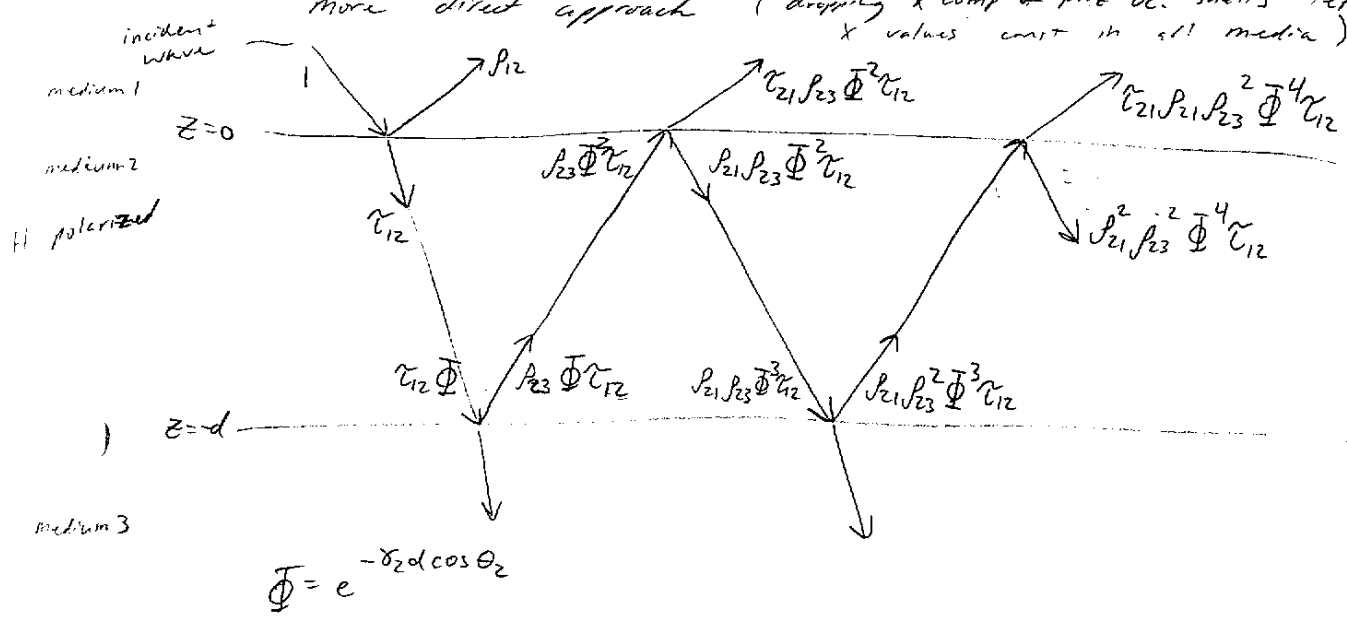
where

$$\rho_{12} = \begin{cases} \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2} & H \text{ pol} \\ \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1} & V \text{ pol} \end{cases}$$

$$\rho_{23} = \begin{cases} \frac{\eta_3 \cos \theta_2 - \eta_2 \cos \theta_3}{\eta_3 \cos \theta_2 + \eta_2 \cos \theta_3} & H \text{ pol} \\ \frac{\eta_3 \cos \theta_3 - \eta_2 \cos \theta_2}{\eta_3 \cos \theta_3 + \eta_2 \cos \theta_2} & V \text{ pol} \end{cases}$$

Fresnel ref coeffs

- But, now that we've laid groundwork, let's take a more direct approach (dropping x comp of phz bc. Snell's tells us that x values exist in all media)



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Ret from 2 layer comp (cont.)

• Now, summing all upward-going waves in medium 1

$$\begin{aligned} \rho &= \rho_{12} + \tilde{r}_{21} \rho_{23} \bar{\Phi}^2 \tilde{r}_{12} + \tilde{r}_{21} \rho_{21} \rho_{23}^2 \bar{\Phi}^4 \tilde{r}_{12} + \dots \\ &= \rho_{12} + \tilde{r}_{12} \tilde{r}_{21} \rho_{23} \bar{\Phi}^2 \sum_{n=0}^{\infty} \chi^n \end{aligned}$$

where

$$\chi = \rho_{21} \rho_{23} \bar{\Phi}^2$$

Note that $|\rho_{21}| < 1$ and $|\rho_{23}| < 1$ and $|\bar{\Phi}| < 1$
 $\Rightarrow |\chi| < 1$

Also

$$\rho_{21} = -\rho_{12} \quad \rightarrow \quad \tilde{r}_{21} = 1 + \rho_{21} = 1 - \rho_{12}$$

Recall that

$$\tilde{r}_{12} = 1 + \rho_{12}$$

Use identity (Taylor series)

$$\sum_{n=0}^{\infty} \chi^n = \frac{1}{1-\chi}$$

$$\Rightarrow \rho = \rho_{12} + \frac{(1+\rho_{12})(1-\rho_{12})\rho_{23}\bar{\Phi}^2}{1 + \rho_{12}\rho_{23}\bar{\Phi}^2}$$

$$\Rightarrow \rho = \frac{\rho_{12} + \rho_{23} e^{-2\delta_2 d \cos \theta_2}}{1 + \rho_{12} \rho_{23} e^{-2\delta_2 d \cos \theta_2}}$$

• Implications:

oscillatory

1. δ_2 is complex valued, so for $d \gtrsim \lambda_2$, $|\rho|$ should have ~~date~~ ^{behavior} w/ max corresponding to constructive interference of multiple reflections and min $\rightarrow$ destructive interference
2. for $d \gg \delta_2$, $\rho \rightarrow \rho_{12}$
3. for $\rho_{23} \ll 1$, $\rho \rightarrow \rho_{12}$